Loïc Strullu 2023

Practical Work 2: Effect of input files on model outputs

We will use the USM that we created in the practical work 1. For memory, we simulate a winter wheat following a spring pea crop at Mons in the Picardie region in Northern France. The simulation begins in summer the 22nd July 2010 after harvest of spring pea in 21st July 2010. The soil is a deep silt-loam soil.

I. Analysis of simulation results

1. Open JavaSTICS and select the workspace Practical_Work_2
2. Launch the simulation Wheat_2011 (reference)
3. Analysis of the simulation balance
4. Comparison of observed and simulated data: 1) biomass and yield (select masecnp and mafruit), 2) nitrate in the soil (select aznit 1 to 4 and azomes)

II. Modification of the initialization file

Illustration of the impact of initialization on model simulations. During the practical work 1 we initialized the model with default values for nitrate and water content. We saw during the comparison between simulated and observed data that nitrate content in the soil was underestimated by the model. After calibration, we determined new values for model initialization concerning the nitrate content in the different soil layers.

1. Create a new initialization file (wheat_2011_new_ini.xml) and modify the nitrate content in the different horizons:

Horizons	NO3initf
1	6 *(instead of 5)*
2	3 *(instead of 5)*
3	7 *(instead of 5)*
4	48 *(instead of 15)*
5	40 *(instead of 15)*

2. Create and launch the simulation Wheat_2011_ini with this new initialization file
3. Comparison of the balance files in JavaSTICS (mod_bWheat_2011 and mod_bWheat_2011_ini): what is modified?
4. Comparison of observed and simulated data: is it better?

III. Impact of climate on simulation

Illustration of the impact of climate on model simulations.

1. Create a new USM Wheat_2020 with climatic data Mons.2019 and Mons.2020

2. Launch the simulation Wheat_2020

5. Comparison of the balance files in JavaSTICS (mod_bWheat_2011 and mod_bWheat_2020): what is modified?

IV. Create a new soil file with variable organic N content

Illustration of the impact of soil organic N content parametrization on model outputs.

Organic N content (Norg) in soil play a major role in the soil mineralization rate. We initialize the soil Norg at 0.105 %, however this value can vary in function of cultural practices. We are going to explore the impact of this parameter on soil N mineralization and on crop growth and yield.

1. Create new soil file from the Mons soil file:
   - ✓ Mons_highN: Norg = 0.13 %
   - ✓ Mons_low_N: Norg = 0.09 %

2. Create new USMs from the USM Wheat_2020 with the corresponding soil file:
   - ✓ Wheat_2020_highN
   - ✓ Wheat_2020_lowN

3. Run the 2 USMs: Wheat_2020_lowN and Wheat_2020_highN

4. Comparison of the simulation balance (Wheat_2020 vs Wheat_2020_lowN and Wheat_2020_highN). What is modified?

V. Create a new technical file for fertilization

Illustration of the impact of fertilization practices on simulations.

1. Create a new technical file called Wheat_fert_2011 from the Wheat_2011 technical file:
   - ✓ Modify the type of fertilizer used by ammonium nitrate

2. Create a new USM Wheat_2011_fert with the corresponding technical file from the Wheat_2011 USM

3. Run the new USM

4. Comparison of the simulation balance compared to the previous USM (Wheat_2011_fert vs Wheat_2011). What is modified?

VI. Create a new technical file to modify the variety used

Illustration of the impact of crop variety on simulations.

1. Create a new technical file called Wheat_2011_var from the Wheat_2011 technical file:
   - ✓ Modify the variety by choosing the variety number one.

2. Create a new USM Wheat_2011_var with the corresponding technical file from the Wheat_2011 USM

3. Run the new USM

4. Comparison of the simulation balance. What is modified ?

VII. Soil depth and irrigation

Illustration of the impact of soil depth and irrigation.

1. Create a new soil file called Mons_sup and modify the maximum root depth (obstarac) at 60 cm

2. Create a new USM Wheat_2011_sup with the corresponding soil file from the Wheat_2011 USM

3. Run the new USM

4. Comparison of the simulations balance Wheat_2011_sup vs Wheat_2011

NB for teachers: Here you can show the comparison between the two balance files.

1) *Input data: modification of the soil*
2) *Crop development: LAI at LAX stage is the same but senescence (LAN stage) is latter*
3) *Growth and components of yield: increase of biomass production and yield. Show the relationship between grains number and growth rate. There is also an increase of crop N content.*
4) *Water and nitrogen balance, show the difference of water stress indices*

5. Create a new technical file called Wheat_IRR_2011 from the Wheat_2011 technical file:
   - ✓ Add water inputs of 40 mm at julian day 505 and 515
     NB for teachers: show how to do it and speak about the different options

6. Create a new USM Wheat_2011_sup_IRR with the corresponding technical file from the Wheat_2011_sup USM

7. Run the new USM

8. Comparison of the simulations balance Wheat_2011_sup vs Wheat_2011_sup_IRR

NB for teachers: Here you can show the comparison between the two balance files.

1) *Input data: modification of the technical file with irrigation inputs*
2) *Crop development: LAI at LAX stage is the same (1ère irrigation au stade LAX ^^), senescence (LAN stage) is latter in the season (6th of Jully vs 7th of June)*
3) *Growth and components of yield: increase of biomass production (+ 7 t ha^{-1}) and yield (+2.8 t ha^{-1}). Show the relationship between grains number (around x2) and growth rate (around x2).*
4) *Water and nitrogen balance, show the difference of water stress indices*
5) *Water, nitrogen and carbon balance:*
   a. *Normalized days for mineralization increased by irrigation*
   b. *Plant N balance: Increased N uptake by the crop (190 vs 122)*
   c. *Soil mineral N balance: increased humus mineralization (146 vs 127) and lower nitrate in the soil at the end of the simulation*